II-Probabilistic Approaches Causality

Foundations of Neuro-Symbolic AI

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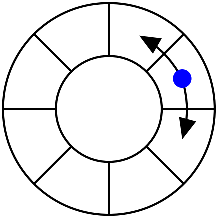
Outline

- Ambiguity in Bayesian Networks
- Modelling interventions
- Reasoning about hypothetical worlds: Counterfactuals

Example: Random walk

Imagine a random walk through circular cells:

- X_k describes at timestep $k \in [d]$ the position
- With probability 0.5 we stay at a cell, and to each neighboring cell with 0.25



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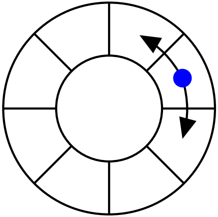

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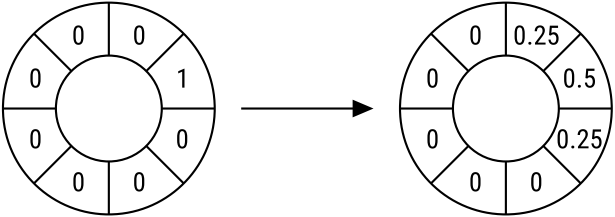
Example: When observing the state 1 for X_k we transit as



$\mathbb{P}[X_k] = \epsilon_1[X_k]$

$\mathbb{P}[X_{k+1}] =$

$$\begin{pmatrix} 0.25 & 0.5 & 0.25 & 0 & 0 & 0 & 0 & 0 \end{pmatrix} [X_{k+1}]$$

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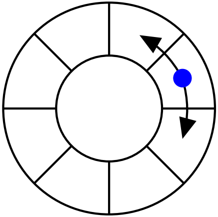

- X_k describes at timestep $k \in [d]$ the position
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This is a Markov chain with the transition matrix:

$$\mathbb{P}\left[X_{k+1} \middle| X_k\right] = \begin{pmatrix} 0.50 & 0.25 & 0 & 0 & 0 & 0 & 0 & 0.25 \\ 0.25 & 0.50 & 0.25 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0.25 & 0.50 & 0.25 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0.25 & 0.50 & 0.25 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.25 & 0.50 & 0.25 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.25 & 0.50 & 0.25 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0.25 & 0.50 & 0.25 \\ 0.25 & 0 & 0 & 0 & 0 & 0 & 0.25 & 0.50 \end{pmatrix} [X_{k+1}, X_k]$$

Reasoning about the future and the past

Intuitive: Conditioning on the past we reason on the future:

$$\left\langle \epsilon_{x_k}\left[X_k\right], \mathbb{P}\left[X_{k+1} \middle| X_k\right] \right\rangle_{[X_{k+1}]} = 0.5 \cdot \epsilon_{x_k}\left[X_{k+1}\right] + 0.25 \cdot \epsilon_{x_k-1}\left[X_{k+1}\right] + 0.25 \cdot \epsilon_{x_k+1}\left[X_{k+1}\right]$$

Counter-intuitive: Conditioning on the future we can also reason on the past:

$$\left\langle \mathbb{P}\left[X_{k+1} \middle| X_k\right], \epsilon_{x_k}\left[X_{k+1}\right] \right\rangle_{[X_k]} = 0.5 \cdot \epsilon_{x_k}\left[X_k\right] + 0.25 \cdot \epsilon_{x_k-1}\left[X_k\right] + 0.25 \cdot \epsilon_{x_k+1}\left[X_k\right]$$

How to distinguish between these directions?

How to distinguish the past from the present?

We have two decomposition schemes as Bayesian networks, both decorated with the same transition matrices:

Parametrization from the past to the future:

$$\cdots \longrightarrow X_{k-2} \longrightarrow X_{k-1} \longrightarrow X_k \longrightarrow X_{k+1} \longrightarrow X_{k+2} \longrightarrow \cdots$$

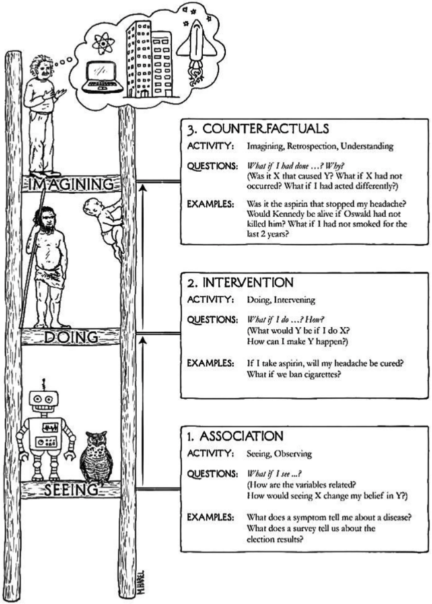
Parametrization from the future to the past:

$$\cdots \longleftarrow X_{k-2} \longleftarrow X_{k-1} \longleftarrow X_k \longleftarrow X_{k+1} \longleftarrow X_{k+2} \longleftarrow \cdots$$

Ambiguity in Bayesian Networks

Causation is not (always) reflected in the directions.

The Ladder of Causality



→ Model randomness by **response variables** L_k and add **counterfactual variables** X_k

→ Model interventions by **do variables** D_k

→ Bayesian Networks model the distribution of **observed variables** X_k

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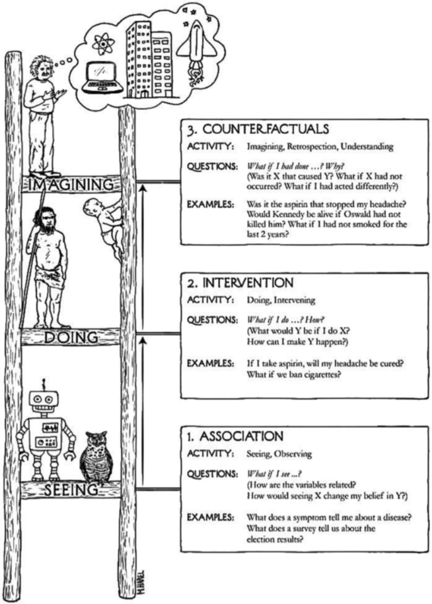

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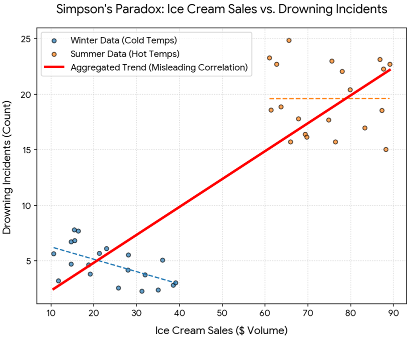
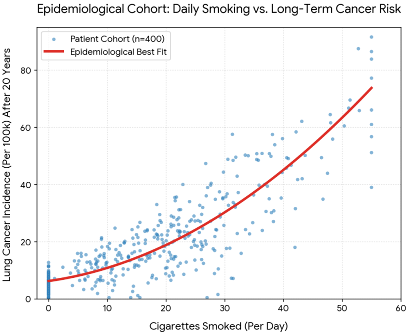
Correlation does not imply causation

Consider two examples[1] of correlations:





Question: How to reduce casualties based on public laws and taxes?

[1] generated data

Interventions

Interventional reasoning

AI is not only about modelling the observed, but reasoning about consequences: When we intervene on a variable, what would be the effect on the further variables?

This appears in many realistic

- Will product sales increase if we lower the price?
- Will I pass the exam tomorrow if I study today?

Idea:

- Understand the directed hyperedges to model causal consequences.
- But: How to deal with the ambiguity?

Associations in Bayesian Networks can not model interventions!

Do variables: Modelling interventions

We add to each variable X_k a do variable D_k of dimension $[m_k + 1]$ modelling the invervention on it:

- Intervention on the variable X_k when $d_k \in [m_k]$
- No intervention on the variable when $d_k = m_k$

This is realized by a tensor

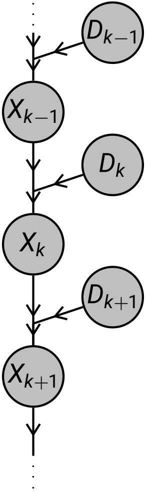
$$\mathbb{P}^c\left[X_k \middle| X_{\mathrm{Pa}(k)}, D_k\right] = \sum_{x_k \in [m_k]} \epsilon_{x_k}\left[X_k\right] \otimes \mathbb{I}\left[X_{\mathrm{Pa}(k)}\right] \otimes \epsilon_{x_k}\left[D_k\right] \quad \text{(intervention)}$$

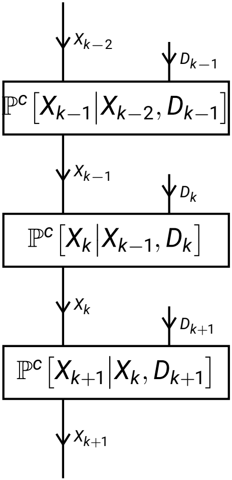
$$+ \mathbb{P}\left[X_k \middle| X_{\mathrm{Pa}(k)}\right] \otimes \epsilon_{m_k}\left[D_k\right] \quad \text{(no intervention)}$$

This resolves the ambiguity in Bayesian network representation!

Causally augmented Markov chain

For a Markov chain we get the causally augmented model:





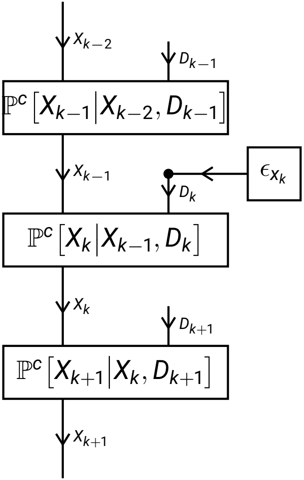
Example: Interventions on a random walk

The causally augmented conditional distributions are for the random walk with $m = 3$:

$$\mathbb{P}^c\left[X_k \middle| X_{k-1}, D_k\right] = \underbrace{\begin{pmatrix}1&1&1\\0&0&0\\0&0&0\end{pmatrix}^{\begin{pmatrix}0&0&0\\1&1&1\\0&0&0\end{pmatrix}^{\begin{pmatrix}0&0&0\\0&0&0\\1&1&1\end{pmatrix}^{\overbrace{\begin{pmatrix}0.50&0.25&0.25\\0.25&0.50&0.25\\0.25&0.25&0.50\end{pmatrix}}^{\textbf{No intervention}}}}}}_{\textbf{Intervention}} [X_k, X_{k-1}, D_k]$$

Intervening on a position

Query: What is the consequence of intervening at time k in state x_k?



Interpretation: Restart of the random walk at time k in state x_k.

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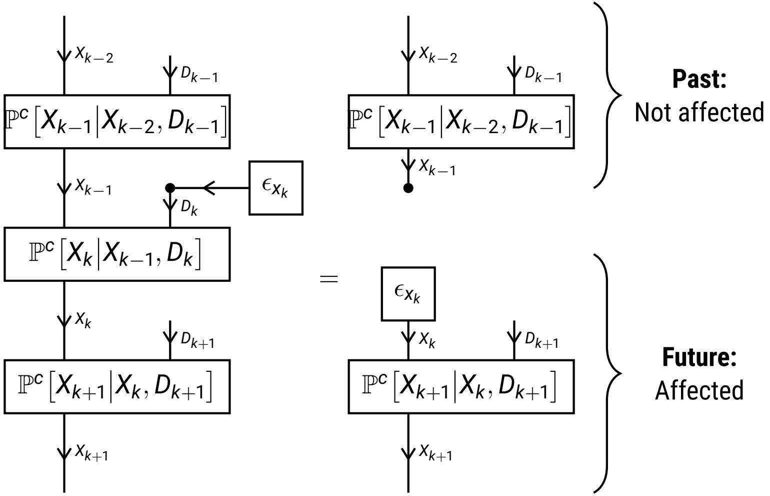


Interpretation: Restart of the random walk at time k in state x_k.

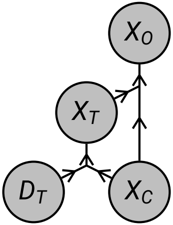
Randomized controlled trial

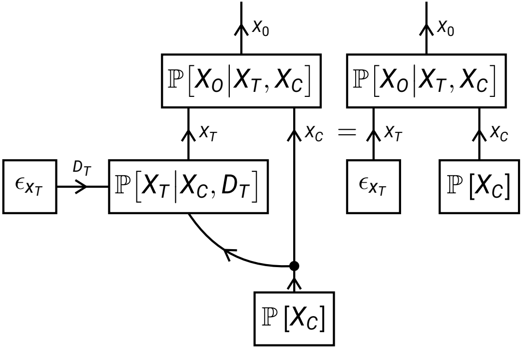
Aim

Model the causal effect of a treatment X_T (e.g. a drug) on an observation X_O (e.g. patients health).

- How to assign the treatment to the patients?
- How to determine the causal effect?

Naive approach: Voluntarily treatment and observation of the effects.





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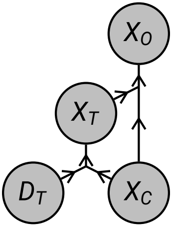

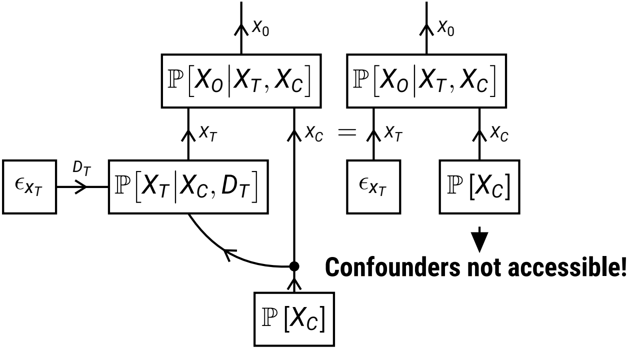

Randomized controlled trial

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- How to assign the treatment to the patients?
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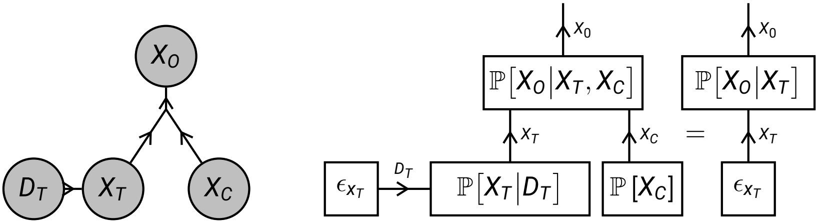
Randomized controlled trial: Assign treatment independently to test group



Influence is identifiable by conditional queries!

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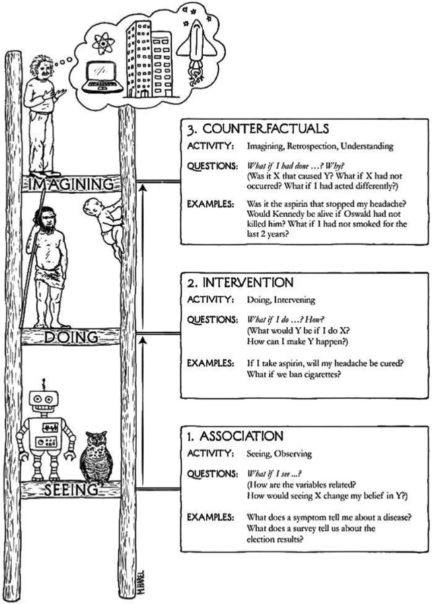

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Counterfactuals

Counterfactual reasoning

- A particular state has been observed.
- We reason about a hypothetical world.

This appears in important situation:

- Would the accident have happened had the safety protocol been respected?
- Would the person have developed cancer had he stopped smoking earlier?

Interventions are not enough for these questions!

Response variables

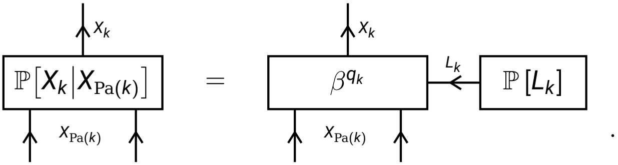
Interpretation of randomness

Random behavior in observed variables is due to exogenous variables, which states are not observed.

To decompose each conditional distribution we introduce:

- Response variables L_k summarize the exogenous variables
- Function q_k describing a deterministic dependence of x_k on the parent states $x_{\mathrm{Pa}(k)}$ and the state ℓ_k of the response variable

The conditional probabilities are then contracted basis encodings, that is



Random walk example

We model a response variable

- $\ell_k = 0$: going to the left cell
- $\ell_k = 0$: staying at the current cell
- $\ell_k = 2$: going to the right cell

$$\mathbb{P}\left[X_k | X_{k-1}, L_k\right] = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}^{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^{\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix}}} \left[X_k, X_{k-1}, L_k\right]$$

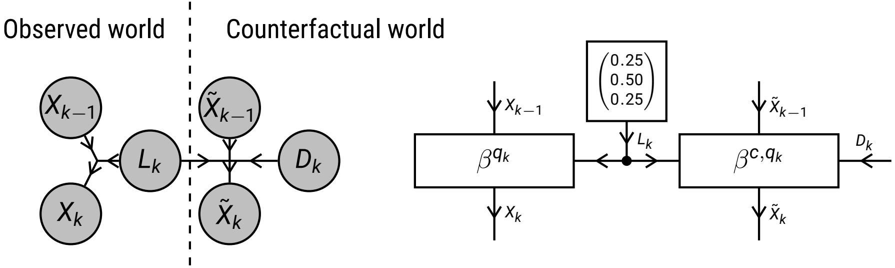
Effect in random walk: Shift of the random walk.

Counterfactual variables

Total setup to reason about interventions on a hypothetical world:

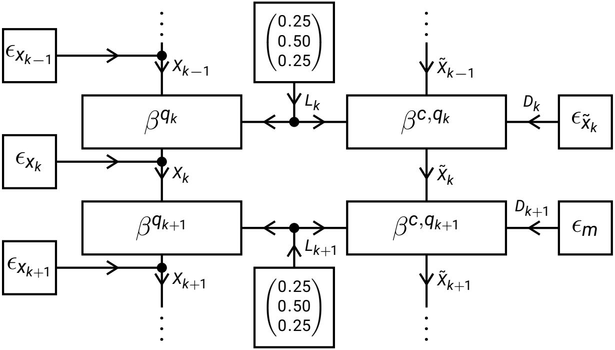
- Observed variables X_k: describing the observed world
- Response variables L_k: summarizing exogenous variables
- Counterfactual variables $\tilde{X}_k$: describing an hypothetic world
- Do variables D_k: interventions in the hypothetic world

In the random walk example:



Counterfactual query

What would be the outcome of the walk, would we have done the intervention on X_k?



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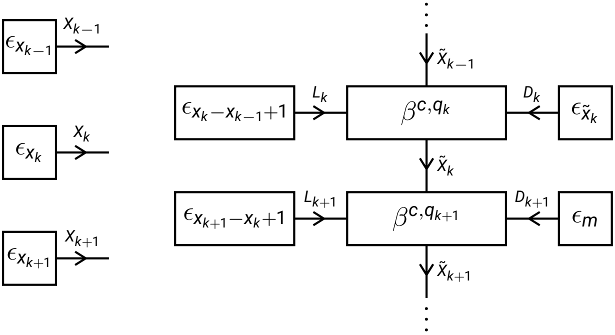

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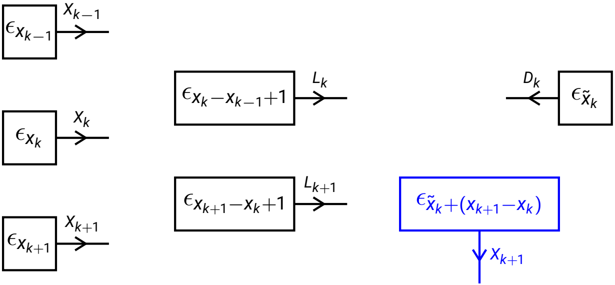

Answer: The walk would be shifted by the intervention. Different to the intervention query, we have a deterministic outcome!